

Data Card: Irokobot Mental Health Support Dataset

Challenge 2 Submission — Dataset Design & Documentation

1. Team & Project Details

Field	Detail
Project Name	Irokobot
Repository	github.com/Ololade117/Iroko
Team Name	Iroko
Submission Date	28/08/2026

Team Members

Name	Role / Contribution
Ezekiel Oladejo	Iroko team member
Habib Bashir Lawal	Iroko team member (Team Lead II)
Ololade Ogunleye	Iroko team member (Team Lead)
Rapheal Aleoh	Iroko team member
Reuben Pometsey	Iroko team member
Saqlain Ahmed P	Iroko team member
Stephanie Omolu	Iroko team member

2. Dataset Overview

Irokobot (Iroko-Med-Bot) responds to a documented gap in Nigeria's mental health system: the country has consistently spent under 5% of its federal budget on healthcare (2015-2025), has only 262 psychiatrists for a population of roughly 240 million, and over 90% of Nigerians have no health insurance. Existing large AI systems are not a safe substitute — research cited in the project's problem statement found that general-purpose AI agrees with harmful suggestions in nearly a third of simulated mental health crises, and can misread culturally-specific expressions of distress as clinical delusions.

Irokobot is designed as a stigma-free, first-line information and triage tool for individuals and families concerned about mental health in Nigeria, with a longer-term project vision of also supporting community health workers as an assistive clinical-support tool. Its purpose is to

provide supportive, evidence-based psychoeducation and emotional support, and to encourage users toward professional help when appropriate, it is explicitly not designed to diagnose or treat mental health conditions.

The dataset documented in this Data Card has two components that work together:

Component	Purpose
Fine-tuning dataset	Instruction/response conversational examples used to train the model's conversational behavior, tone, and safety responses.
RAG knowledge base	Chunked reference passages retrieved at question-time to ground the model's factual answers in real source material, so answers can be traced back to a source rather than functioning as an untraceable "black box."

3. Input vs. Output

	Description
Input	Free-text messages typed by the user (patient), in English. Each input may optionally be accompanied by prior conversation history, allowing the model to respond with awareness of context across a multi-turn conversation.
Output	A free-text supportive or psychoeducational response generated by the fine-tuned model. Where relevant, the response is grounded in reference passages retrieved from the RAG knowledge base rather than generated purely from the model's trained parameters.
Modality	Text only. No image, audio, or video data is used or produced.

4. Classification Task

The RAG knowledge base is organized into six topic categories, used to route retrieval rather than to classify or diagnose the user:

Category	Type	Rationale
Depression	Informational	High prevalence; well-documented in public health sources.

Anxiety	Informational	High prevalence; well-documented in public health sources.
PTSD	Informational	Well-documented; strong public educational content available.
Bipolar Disorder	Informational	Commonly discussed condition with accessible public information.
OCD	Informational	Commonly discussed condition with accessible public information.
Triage	Behavioral	Not a diagnosis: governs how the model responds to crisis-adjacent or urgent messages (calm de-escalation and redirection to real crisis resources), kept separate from informational content.

Schizophrenia and eating disorders were deliberately excluded from this initial dataset's scope, despite schizophrenia being cited prominently in the project's problem statement (affecting an estimated 24 million people worldwide, with particular relevance to Nigeria given late average diagnosis age and high stigma). This is a genuine, acknowledged gap rather than a deliberate long-term exclusion: schizophrenia involves higher-risk symptom presentations (e.g. psychosis) that need more clinical oversight than the current phase's psychoeducation-focused dataset provides. Eating disorder content carries specific safe-messaging requirements (e.g. avoiding numeric/specific detail) that add complexity beyond the current scope. Both are flagged as priorities for a future phase rather than permanent exclusions, given their centrality to the project's stated motivation.

5. Labelling Strategy

No new manual labelling was performed by the team on the core conversational data. Labels and structure were inherited from the source datasets:

Source	Labelling origin
CounselChat	Therapist-authored responses to real user questions, organized by topic as published in the original dataset.
Mental Health FAQ	Pre-compiled question/answer pairs, no additional labelling applied.
Mental Health Conversations	Pre-compiled context/response pairs, no additional labelling applied.

ESConv	Emotion type, problem type, and support-strategy annotations applied by the original dataset's academic authors (Tsinghua University).
AnnoMI	Motivational-interviewing quality and strategy annotations applied by trained annotators as part of the original academic study (University College Cork).

The 17 newly authored safety examples (identity disclosure and refusal/redirection responses) were drafted directly by the team to address a gap identified during a safety audit, and were reviewed and approved by the team lead before merging into the training set.

Quality control: a rule-based (regex pattern-matching) safety audit was run across the full dataset to flag potential issues — fabricated identity/credential claims and non-Nigeria regional references. Every flagged row was manually reviewed before being edited, ensuring automated changes did not silently corrupt valid responses. One row that could not be safely auto-edited was flagged for manual correction rather than shipped as-is.

6. Data Creation

Fine-tuning dataset sources:

Dataset	Type	Origin
CounselChat	Single-turn Q&A	Public dataset (Kaggle), sourced from a public online counseling platform.
Mental Health FAQ	Single-turn Q&A	Public dataset (Kaggle).
Mental Health Conversations	Single-turn Q&A	Public dataset (Kaggle).
ESConv	Multi-turn dialogue	Public academic dataset (GitHub, Tsinghua University).
AnnoMI	Multi-turn dialogue	Public academic dataset (GitHub, University College Cork).

RAG knowledge base sources:

Reference text scraped from official public health organization webpages: the U.S. National Institute of Mental Health (NIMH), the National Center for PTSD, the International OCD Foundation, and SAMHSA/988 Suicide & Crisis Lifeline. Pages were chunked into approximately 120-word passages with overlap, to support accurate retrieval at question-time.

Explicit exclusion: a Reddit-sourced depression dataset was deliberately excluded from the fine-tuning data, to avoid training the model on unverified or potentially harmful community-generated responses.

Processing pipeline: cleaning (removing null/duplicate/too-short rows), schema unification into a common instruction/input/output format, deduplication, merging of single-turn and multi-turn sources, a safety audit and correction pass (see Section 8), and an 80/10/10 train/validation/test split. The final dataset used for training contains 7,616 examples.

7. Web Collection Ethics

RAG source pages were collected exclusively from official public health bodies (NIMH, National Center for PTSD, IOCDF, SAMHSA) that publish free, publicly accessible health information intended for public education and reuse. No login-gated, paywalled, or personal content was accessed. No personal data was collected during this process.

The Kaggle-sourced conversational datasets (CounselChat, Mental Health FAQ, Mental Health Conversations) were used as pre-compiled public datasets rather than scraped directly by the team. Dataset license terms were checked prior to use; datasets with restrictive redistribution terms were excluded from any public-facing repository push.

Privacy consideration: a safety audit specifically identified and removed instances where real, identifiable professional details (e.g. a named therapist and their workplace) had leaked from the original CounselChat source data into training responses. This prevents the model from reproducing or fabricating identifiable personal or professional claims about real individuals.

8. Bias and Representation

Identified gap	Description / Mitigation
Geographic/cultural mismatch	Source data is predominantly Western (U.S.-based platforms and institutions), while the intended user base is Nigerian. Crisis resources embedded in training responses were audited and replaced with Nigeria-specific resources (SURPIN toll-free helpline, Mentally Aware Nigeria Initiative, and the 112 national emergency line) wherever U.S./Canada/UK references were found.
Language coverage	The dataset is entirely in English. No Nigerian Pidgin, Yoruba, Hausa, or Igbo content is currently included, despite wide use of these languages within the target community. Identified as ongoing work.
Condition coverage imbalance	Some categories (e.g. OCD) have substantially fewer source examples than higher-

	prevalence categories (e.g. depression, anxiety), which may lead to weaker model performance for lower-resource categories.
Excluded conditions	Schizophrenia and eating disorders are not currently covered, given the additional clinical sensitivity and safe-messaging requirements involved (see Section 4).
Identity/credential fabrication	An audit found 93 responses (from the original CounselChat data) containing fabricated identity or licensing claims (e.g. a response claiming to be a named, licensed therapist). 91 were corrected by removing the fabricated claim; 1 was flagged for manual review rather than auto-edited.
Refusal-behavior gap	An audit for harmful-compliance risk found no existing bad examples in the dataset, but confirmed an absence of training examples teaching the model to refuse violent or self-harm-adjacent requests. 17 new examples were authored to begin closing this gap; the team considers this a first-draft seed set requiring further expansion proportional to real-world query volume.

9. Values and Community Alignment

Challenge 1 problem statement (summary): Nigeria's mental health system is severely under-resourced — under 5% of the federal budget goes to healthcare, there are only 262 psychiatrists for roughly 240 million people, and over 90% of Nigerians have no health insurance. General-purpose AI systems are not a safe substitute: cited research found they agree with harmful suggestions in nearly a third of simulated mental health crises and misread culturally-specific expressions of distress as clinical delusions, while remaining "black boxes" that clinicians don't trust. Iroko-Med-Bot's stated goal is a stigma-free, culturally-safe, source-traceable SLM that serves as a first-line triage and information tool for individuals and families, with a longer-term vision of also supporting community health workers.

Key values guiding dataset design, drawn directly from that problem statement:

Value	How the dataset reflects it
Stigma-free support	Fine-tuning data was audited to remove fabricated authority claims and reviewed for tone, aiming for responses that inform and support rather than judge.
Cultural safety	Identified as a current gap rather than a solved problem: source data is

	Western/English-centric, and the project's own research basis warns against misreading culturally-specific distress as pathology. Localizing crisis resources to Nigeria (SURPIN, MANI, 112) is a first step; broader cultural and language adaptation is ongoing work (Section 10).
Traceable, non-"black box" answers	The RAG architecture was specifically chosen so factual answers are grounded in retrievable source passages rather than opaque model memorization, directly addressing the clinician-trust problem named in the problem statement.
Avoids agreeing with harmful suggestions	A dedicated safety audit targeted this exact failure mode; new refusal-and-redirect training examples were authored for self-harm-adjacent and violence-adjacent prompts.
Triage over diagnosis	Dataset scope is limited to psychoeducation, emotional support, and appropriate escalation — not diagnosis or treatment — consistent with the project's framing as a first-line triage tool, not a replacement for the 262 psychiatrists serving the country.
Accountability	Every safety-related dataset edit is tracked in a change log documenting original and corrected text, rather than silently altering data. Known gaps (schizophrenia coverage, language localization) are documented rather than hidden.

10. Known Limitations & Ongoing Work

- Schizophrenia coverage is a documented gap relative to the project's own problem statement, which specifically motivates the project using schizophrenia prevalence and stigma in Nigeria. Adding this category requires additional clinical-safety design work before inclusion.
- RAG knowledge base content is not yet localized to Nigeria — current sources are U.S.-based public health institutions.
- No native-language (Pidgin, Yoruba, Hausa, Igbo) support is currently included, despite the problem statement's stated goal of training on local languages.
- The dataset currently supports direct individual/family conversational use; the problem statement's longer-term vision of a community-health-worker clinical support tool would likely need additional, more clinically-oriented data not yet sourced.

- The 17-example safety/refusal training set is a first draft, not yet proportional to real-world adversarial query volume.
- The team plans to mix a portion of the base model's own original safety-aligned instruction data into fine-tuning, to reduce the risk of the model losing prior safety alignment during training ("catastrophic forgetting") this data has not yet been sourced.
- The dataset reflects pre-deployment data only; no real-user feedback loop currently informs the dataset.

11. Dataset Statistics Summary

Metric	Value
Total fine-tuning examples	7,616
Single-turn source datasets	3 (CounselChat, Mental Health FAQ, Mental Health Conversations)
Multi-turn source datasets	2 (ESConv, AnnoMI)
RAG knowledge base categories	6 (Depression, Anxiety, PTSD, Bipolar Disorder, OCD, Triage)
Rows corrected in safety audit	91 (identity claims and/or non-Nigeria region references)
Rows flagged for manual review	1
New safety/refusal training examples added	17